Basic Blood, Diabetes & Lipid Tests (20)
(Core Blood & Metabolic Tests)

- Complete Blood Count (CBC)
- Hemoglobin (Hb)
- Red Blood Cell Count (RBC)
- White Blood Cell Count (WBC)
- Platelet Count
- Hematocrit (HCT)
- Mean Corpuscular Volume (MCV)
- Mean Corpuscular Hemoglobin (MCH) [Present within RBC Indices Test]
- Mean Corpuscular Hemoglobin Concentration (MCHC) [Present within RBC Indices Test]
- Erythrocyte Sedimentation Rate (ESR)
- Fasting Blood Sugar (FBS) [Present within Blood Glucose Test]
- Postprandial Blood Sugar (PPBS)
- Random Blood Sugar (RBS) [Present within Diabetes test]
- Hemoglobin A1c (HbA1c)
- Insulin Test
- Total Cholesterol
- LDL Cholesterol
- HDL Cholesterol
- Triglycerides [Present within cholesterol test.]
- VLDL Cholesterol

Complete Blood Count (CBC)

What is a Complete Blood Count?

A complete blood count, or CBC, is a group of blood tests that measure the number and size of the different cells in your blood. A CBC measures:

- **Red blood cells**, which carry oxygen from your lungs to the rest of your body.
- **White blood cells**, which fight infections and other diseases. There are five major types of white blood cells. A CBC test measures the total number of white cells in your blood. A different test called a **CBC with differential** measures the number of each type of these white blood cells.
- **Platelets**, which stop bleeding by helping your blood to clot.
- **Hemoglobin**, an iron-rich protein in red blood cells that carries oxygen from your lungs to the rest of your body.
- **Hematocrit**, the amount of whole blood that is made up of red blood cells.
- **Mean corpuscular volume (MCV)**, the average size of your red blood cells.

Other names for a complete blood count: CBC, full blood count, blood cell count

What is it used for?

A complete blood count is often part of a routine checkup. It is also used to monitor a condition or treatment that may affect your blood cell counts such as infections, anemia, immune system disorders, and blood cancers.

Why do I need a complete blood count?

Your health care provider may have ordered a complete blood count as part of your checkup or to monitor your overall health. The test may also be used to:

- Help diagnose blood diseases, infections, immune system disorders, or other medical conditions
- Check for changes in an existing blood disorder

What happens during a complete blood count?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

Usually, no special preparation is necessary for a complete blood count. But if your provider ordered other tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

Are there any risks to the test?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle went in, but most symptoms go away quickly.

What do the results mean?

A CBC counts the cells in your blood. There are many reasons your levels **may not be in the normal range**. For example:

- Abnormal levels of red blood cells, hemoglobin, or hematocrit may be a sign of dehydration, anemia, heart disease, or too little iron in your body.
- A low white cell count may be a sign of an autoimmune disorder, bone marrow disorder, or cancer.
- A high white cell count may be a sign of an infection or a reaction to a medicine.

If any of your levels are abnormal, it doesn't always mean you have a medical condition that needs treatment. Diet, activity level, medicines, having a menstrual period, not drinking enough water, and other factors can affect the results. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

Is there anything else I need to know about a complete blood count?

A complete blood count is only one tool your provider uses to learn about your health. Your provider will consider your medical history, symptoms, and other factors to make a diagnosis. You may also need additional tests.

Hemoglobin Test

What is a Hemoglobin Test?

A hemoglobin test measures the levels of hemoglobin in your blood. Hemoglobin is an iron-rich protein in your red blood cells. It carries oxygen from your lungs to the rest of your body. Your cells need oxygen to grow, reproduce, and make energy for you to function. Your hemoglobin levels provide information about the health of your red blood cells. If your hemoglobin levels are abnormal, it may be a sign that you have a blood disorder.

Other names: Hb, Hgb, H&H

What is it used for?

A hemoglobin test is a blood test often used to check for anemia, a condition in which your body does not make enough healthy red blood cells. If you have anemia, the cells in your body don't get all the oxygen they need. Hemoglobin levels are usually measured as part of a complete blood count (CBC). A CBC is a group of tests that measure the number and type of cells in your blood.

A hemoglobin test is not the same as a hemoglobin A1C test. The A1C test is used to check your average blood glucose (blood sugar).

Why do I need a hemoglobin test?

Your health care provider may order the test as part of a routine exam, or if you have:

- Symptoms of anemia, which include weakness, dizziness, and cold hands and feet
- A family history of thalassemia, sickle cell anemia, or other inherited blood disorder
- A diet low in iron and other minerals
- A long-term infection
- Excessive blood loss from an injury or surgical procedure

You may also need this test to monitor the treatment of anemia or the other health conditions that could affect your hemoglobin level.

What happens during a hemoglobin test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

You don't need any special preparation for a hemoglobin test. If your provider has ordered other blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

Your results will compare your hemoglobin level to the normal range, which may be called the reference range. This range may vary, based on the laboratory used and your age, race, and sex. The test can't tell you what's causing an abnormal level. But knowing how high or low your hemoglobin level is can help your provider determine how severe your condition may be.

Your hemoglobin levels may not be in the reference range for many reasons.

Low hemoglobin levels may be a sign of:

- Different types of anemia
- Thalassemia
- Iron deficiency
- Liver disease
- Cancer and other diseases
- Blood loss from injury, a heavy menstrual period, or other problems
- Lack of certain nutrients, such as iron or vitamin B-12

High hemoglobin levels may be a sign of:

- Lung disease
- Heart disease
- Polycythemia vera, a disease which causes your bone marrow to make too many red blood cells
- Sleep apnea, a disorder that causes you to repeatedly stop breathing during sleep

If any of your levels are abnormal, it doesn't always mean you have a medical condition that needs treatment. Living at a higher altitude can also cause a high hemoglobin level. Other factors that can affect your results can include:

- Diet
- Activity level
- Certain medicines
- Dehydration
- Smoking
- Certain genetic conditions

Talk with your provider if you have questions about your results. To understand your hemoglobin results, your provider will also consider your symptoms, medical history, and the results of other blood tests.

Is there anything else I need to know about a hemoglobin test?

Some forms of anemia are mild, while other types of anemia can be serious and even life threatening if not treated. If you are diagnosed with anemia, talk to your provider about the best treatment plan for you.

Red Blood Cell (RBC) Count

What is a red blood cell (RBC) count?

A red blood cell (RBC) count measures the number of red blood cells, also known as erythrocytes, in your blood. Red blood cells are made in your bone marrow, the spongy tissue inside your large bones. They contain hemoglobin, an iron-rich protein that carries oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy for you to function. An RBC count that is higher or lower than normal is often the first sign of an illness. So, the test may allow you to get treatment even before you have symptoms.

Other names: erythrocyte count, red count, RBC Count

What is it used for?

A red blood cell (RBC) count is almost always part of a complete blood count (CBC), a group of tests that measures the number and type of cells in your blood. The RBC measurement is used to help diagnose red blood cell disorders, such as anemia, a condition in which your body does not make enough healthy red blood cells.

Why do I need a red blood cell count?

You may get this test as part of a CBC, which is often included in a routine checkup. You may also need this test if you are getting treatment that might affect your blood cell counts, such as chemotherapy, to monitor health conditions, such as chronic kidney disease, or if you have symptoms of a low or high red blood cell count.
Symptoms of a low red blood cell count may include:

- Shortness of breath
- Weakness or fatigue
- Headache
- Dizziness
- Arrhythmia (a problem with the rate or rhythm of your heartbeat)
- Pale skin

Symptoms of a high red blood cell count may include:

- Headache
- Dizziness
- Vision problems

What happens during a red blood cell count?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for this test?

You don't need any special preparations for a red blood cell (RBC) count.

Are there any risks to this test?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

Your results will show whether you have a normal red blood cell count or a count that is too low or too high.

Conditions that may cause a low red blood cell count can include:

- Anemia
- Leukemia, a type of blood cancer
- Malnutrition, a condition in which your body does not get the calories, vitamins, and/or minerals needed for good health
- Multiple myeloma, a cancer of the bone marrow
- Kidney failure
- Blood loss from injury or other problems
- Alcohol use disorder (AUD)
- Pregnancy

Conditions that may cause a high red blood cell count can include:

- Dehydration
- Heart disease
- Polycythemia vera, a disease which causes your bone marrow to make too many red blood cells
- Scarring of the lungs, often due to cigarette smoking
- Lung disease
- Kidney cancer
- Sleep apnea, a disorder that causes you to repeatedly stop breathing during sleep

Misuse of certain drugs for athletic performance (such as anabolic steroids) and living at a high altitude can also cause a high red blood cell count.
Talk to your health care provider if you have questions about your results. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand your red blood cell count results.

Is there anything else I need to know about an RBC count?

If results showed you had a low or a high red blood cell count, you may need more tests to help your provider make a diagnosis. These can include:

- Reticulocyte count, which measures the number of reticulocytes in your blood. Reticulocytes are immature (still developing) red blood cells. This test helps check if your bone marrow is making the right amount of red blood cells.
- Iron tests, which measure iron levels in your blood. Iron is essential for making red blood cells.
- Vitamin B test, which measures the amount of one or more B vitamins in your blood. B vitamins are important for making red blood cells and keeping your body healthy.

White Blood Count (WBC)

What is a white blood count (WBC)?

A white blood count measures the number of white blood cells (WBCs) in your blood. White blood cells, also called leukocytes, are part of your immune system. They are a type of blood cell made in your bone marrow and found in your blood and lymph tissue (part of your immune system). If you are injured or get sick, the white blood cells will travel through your bloodstream and tissues to where they are needed. There, they will help your body fight off infections and other diseases.

When you get sick, your body makes more white blood cells to fight the bacteria, viruses, or other foreign substances causing your illness. This increases your white blood count.

Other diseases can cause your body to make fewer white blood cells than you need. This lowers your white blood count. Diseases that can lower your white blood count include some types of cancer and HIV, a viral disease that attacks white blood cells. Certain medicines, including chemotherapy, may also lower the number of your white blood cells.

There are five major types of white blood cells:

- Neutrophils
- Lymphocytes
- Monocytes
- Eosinophils
- Basophils

A white blood count measures the total number of these cells in your blood. Another test, called a blood differential, measures the amount of each type of white blood cell.

Other names: WBC count, white cell count, white blood cell count, Leukocyte Count, WBC

What is it used for?

A white blood count is most often used to help diagnose or monitor disorders related to having a high white blood cell count or low white blood cell count.

Disorders related to having a **high white blood count** include:

- Autoimmune and inflammatory diseases, conditions that cause the immune system to attack healthy tissues
- Bacterial or viral infections
- Cancers such as leukemia and Hodgkin disease
- Allergic reactions

Disorders related to having a **low white blood count** include:

- Diseases of the immune system, such as HIV
- Lymphoma, a cancer of the bone marrow
- Diseases of the liver or spleen

A white blood count can show if the number of your white blood cells is too high or too low, but it can't confirm a diagnosis. So it is usually done along with other tests to help confirm your diagnosis. These other tests could include a complete blood count, blood differential, blood smear, and/or bone marrow test.

Why do I need a white blood count?

You may need this test if you have signs of an infection, inflammation, or autoimmune disease. The symptoms of an infection may include:

- Fever
- Chills
- Body aches
- Headache
- Wound that is red, has pus, or won't heal
- Ongoing cough

The symptoms of inflammation and autoimmune diseases will be different, depending on where in the body the inflammation is and the type of disease you have.

You may also need this test if you have a disease that weakens your immune system or are taking medicine that lowers your immune response. If the test shows your white blood count is getting too low, your provider may want to adjust your treatment.

Your newborn or older child may also be tested as part of a routine screening or if they have symptoms of a white blood cell disorder.

What happens during a white blood count?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

To test children, a health care provider will take a sample from the heel (newborns and young babies) or the fingertip or arm (older babies and children). The provider will clean the heel, fingertip or arm with alcohol and poke the site with a small needle. The provider will collect a blood sample and put a bandage on the site.

Will I need to do anything to prepare for the test?

You don't need any special preparations for a white blood count. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

Are there any risks to the test?

After a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is very little risk to your baby or child with a needle stick test. Your child may feel a little pinch when the site is poked, and a small bruise may form at the site. This should go away quickly.

What do the results mean?

Conditions that may cause a **high white blood count**, also called leukocytosis, include:

- An infection
- An inflammatory disease such as rheumatoid arthritis
- An allergy
- Leukemia or Hodgkin disease
- Tissue damage from a burn injury or surgery

Smoking, stress, a reaction to a medicine, or pregnancy may also cause your body to make more white blood cells, which leads to a high white blood count.

Conditions that may cause a **low white blood count**, also called leukopenia, include:

- Bone marrow damage. This may be caused by infection, disease, or treatments such as chemotherapy.
- Cancers that affect the bone marrow.
- An autoimmune disorder, such as lupus.
- HIV.

If you are already being treated for a white blood cell disorder, your results may show if your treatment is working or whether your condition has improved.

If you have questions about your results, talk to your provider. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand your white blood count results.

Learn more about laboratory tests, reference ranges, and understanding results.

Is there anything else I need to know about a white blood count?

White blood count results are often compared with results of other blood tests, including a blood differential. A blood differential test shows the amount of each type of white blood cell, such as neutrophils or lymphocytes. Neutrophils mostly target bacterial infections. Lymphocytes mostly target viral infections.

- A higher-than-normal amount of neutrophils is known as neutrophilia.
- A lower-than-normal amount is known as neutropenia.
- A higher-than-normal amount of lymphocytes is known as lymphocytosis.
- A lower-than-normal amount is known as lymphopenia.

Platelet Tests

What are platelet tests?

Platelets, also known as thrombocytes, are small blood cells. They form in your bone marrow, the sponge-like tissue in your bones. Platelets are essential for blood clotting. A blood clot is a mass of blood that forms when platelets, proteins, and cells in the blood stick together. When you get hurt, your body forms a blood clot to stop the bleeding.
There are two types of platelet tests: a platelet count test and platelet function tests.

A platelet count test measures the number of platelets in your blood:

- A **lower-than-normal platelet count** is called thrombocytopenia. This condition can cause you to bleed too much after a cut or other injury that causes bleeding.
- A **higher-than-normal platelet count** is called thrombocytosis. This can make your blood clot more than you need it to. Blood clots can be dangerous because they can block blood flow.

Platelet function tests check your platelets' ability to form clots. Platelet function tests may include one or more of the following:

- **Closure time.** This test measures the time it takes for platelets in a blood sample to plug a small hole in a tiny tube. It helps screen for different platelet disorders.
- **Viscoelastometry.** This test measures the strength of a blood clot as it forms. A blood clot has to be strong to stop your bleeding.
- **Platelet aggregometry.** This group of tests is used to measure how well platelets clump together (aggregate).
- **Lumiaggregometry.** This test measures the amount of light produced when certain substances are added to a blood sample. It can help show if there are defects in the platelets.
- **Flow cytometry.** This test uses lasers to look for proteins on the surface of platelets. It can help diagnose inherited platelet disorders. This specialized test is only available at certain hospitals and laboratories.
- **Bleeding time.** This test measures how long it takes for bleeding to stop after small cuts are made in your forearm. It was once commonly used to screen for various platelet disorders but is not used much anymore.

Other names: platelet count, PLT, PLT count thrombocyte count, platelet function tests, PFT, platelet function assay, PFA, platelet aggregation studies

What are they used for?

A platelet count is most often used to monitor or diagnose conditions that cause too much bleeding or clotting. A platelet count may be included in a complete blood count, a test that is often done as part of a regular checkup.
Platelet function tests may be used to:

- Help diagnose certain platelet diseases.
- Check platelet function during complex surgical procedures, such as coronary artery bypass surgery and trauma surgery. These types of procedures have an increased risk of bleeding.
- Check patients before surgery if they have a personal or family history of bleeding disorders.
- Monitor people who are taking blood thinners. These medicines may be given to reduce clotting in people at risk for heart attack or stroke.

Why do I need a platelet test?

You may need platelet count and/or platelet function testing if you have symptoms of having lower-than-normal or higher-than-normal platelets.

Symptoms of **lower-than-normal platelets** include:

- Bleeding for a long time after a minor cut or injury.
- Nosebleeds.
- Unexplained bruising.
- Pinpoint sized red spots on the skin, known as petechiae.
- Purplish, red, or brown spots on your skin, known as purpura. These may be caused by bleeding under the skin.
- Menstrual periods that have a heavy flow or last a long time.
- Blood in your urine, stool (poop), or from your rectum (where stool passes out of your body).

Symptoms of **higher-than-normal platelets** include:

- Numbness of hands and feet
- Headache
- Dizziness
- Weakness
- Pain, swelling, and warmth in your lower legs

You may also need platelet function testing if you are:

- Having a complicated surgery
- Taking medicines to lower blood clotting

What happens during a platelet test?

Most platelet tests are done on a blood sample. During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

You don't need any special preparations for a platelet count test. If you are getting a platelet function test, you may need to stop taking certain medicines, such as aspirin and ibuprofen, before your test. Tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. They will let you know if there are any special instructions to follow.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

If your results show **a lower-than-normal platelet count** (thrombocytopenia), it may be a sign of:

- A cancer that affects the blood, such as leukemia or lymphoma.
- A viral infection, such as mononucleosis, hepatitis, or measles.
- An autoimmune disease. This is a type of disease that causes your body to attack its own healthy tissues, which can include platelets.

- Infection or damage to your bone marrow.
- Cirrhosis.
- Vitamin B12 deficiency (not enough).
- Gestational thrombocytopenia. A common, but mild, low-platelet condition that can occur during pregnancy. It is not known to cause any harm to the mother or fetus. It usually gets better on its own during pregnancy or after birth.

If your results show **a higher-than-normal platelet count** (thrombocytosis), it may be a sign of:

- Certain types of cancer, such as lung cancer or breast cancer
- Anemia
- Inflammatory bowel disease, such as Crohn's disease and ulcerative colitis
- Rheumatoid arthritis
- A viral or bacterial infection
- Severe blood loss

If your platelet function test results were **not normal**, it may be a sign of an inherited or acquired platelet disorder:

- **Inherited disorders** are passed down from your family. These conditions are present at birth, but you may not have symptoms until you are older.
- **Acquired disorders** are not present at birth. They may be caused by other diseases, medicines, or environmental exposure. Sometimes, the cause is unknown.

Inherited platelet disorders include:

- Von Willebrand disease, a genetic disorder that reduces the production of platelets or causes the platelets to work less effectively. It can cause excess bleeding.
- Glanzmann thrombasthenia, a disorder that affects platelets' ability to clump together.
- Bernard-Soulier syndrome, another disorder that affects platelets' ability to clump together.
- Storage pool disease, a condition that affects platelets' ability to release substances that help platelets clump together.

Acquired platelet disorders may be due to chronic diseases such as:

- Kidney failure
- Certain types of leukemia
- Myelodysplastic syndrome (MDS), a disease of the bone marrow

Your provider may consider your symptoms, medical history, and the results of other blood tests to understand the results of your platelet tests.

Is there anything else I need to know about platelet function tests?

Platelet tests are sometimes done along with one or more of the following blood tests:

- MPV blood test, which measures the size of your platelets
- Partial thromboplastin time (PTT) test, which measures the time it takes for your blood to clot
- Prothrombin time and INR test, which measures how many seconds it takes for a clot to form in your blood sample

If you are taking anti-platelet medicine, like aspirin, your results show how well your medicine is working.

Hematocrit Test

What is a hematocrit test?

A hematocrit test is a blood test that measures the amount (percent) of your blood that is made up of red blood cells. Red blood cells carry oxygen from your lungs to the rest of your body. The other parts of your blood include white blood cells (to help fight infection and other diseases), platelets (to help make blood clots to stop bleeding), and a liquid called plasma. Hematocrit levels that are too high or too low can be a sign of a blood disorder, dehydration, or other medical conditions that affect your blood.
Other names: HCT, packed cell volume, PCV, Crit; H and H (Hemoglobin and Hematocrit)

What is it used for?

A hematocrit (HCT) test is often part of a complete blood count (CBC). A CBC is a common blood test that measures the number and type of cells in your blood. It is used to check your general health. It may also be used to monitor or help diagnose blood disorders. These could include anemia, a condition in which you don't have enough red blood cells, and polycythemia (erythrocytosis), an uncommon disorder in which you have too many red blood cells and your blood becomes too thick.

Why do I need a hematocrit test?

Your health care provider may order a hematocrit test as part of your routine checkup or to monitor your health if you are being treated for cancer or have an ongoing health condition. Your provider may also order this test if you have symptoms of a red blood cell disorder, such as anemia or polycythemia:

Symptoms of anemia (too few red blood cells) may include:

- Shortness of breath
- Weakness or fatigue
- Headache
- Dizziness
- Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Symptoms of polycythemia (too many red blood cells) may include:

- Headache
- Feeling light-headed or dizzy
- Shortness of breath when lying down
- Weakness or fatigue
- Skin symptoms such as itching after a shower or bath, burning, or a red face
- Heavy sweating, especially during sleep
- Blurred or double vision and blind spots
- Unusual bleeding, such as bleeding gums or nosebleeds

What happens during a hematocrit test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

You don't need any special preparations for a hematocrit test. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

Are there any risks to the test?

There is very little risk to a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

Your hematocrit test results are reported as a number. That number is the percentage of your blood that's made up of red blood cells. For example, if your hematocrit test result is 42, it means that 42% of your blood is red blood cells, and the rest is made up of white blood cells, platelets, and blood plasma.

Normal hematocrit levels will vary, depending on your sex, age, whether or not you smoke, and the altitude where you live. Ask your provider what your hematocrit level should be.
A lower-than-normal hematocrit level may be a sign that:

- You have had blood loss due to a recent injury
- You have a condition causing chronic (long-term) blood loss
- Your body doesn't have enough red blood cells (anemia). There are many types of anemia, and they can be caused by different medical conditions.
- Your body is making too many white blood cells, which may be caused by:
    - Bone marrow disease
    - Certain cancers, including leukemia, lymphoma, multiple myeloma, or cancers that spread to your bone marrow from other parts of the body

A higher-than-normal hematocrit level may be a sign that:

- Your body is making too many red blood cells, which may be caused by:
    - Lung disease
    - Congenital heart disease
    - Heart failure
    - Polycythemia
- Your blood plasma level is too low, which may be caused by:
    - Dehydration, the most common cause of a high hematocrit
    - Shock

If your results are not in the normal range, it doesn't always mean you have a medical condition that needs treatment. Living at high altitudes with less oxygen in the air may cause a high hematocrit. That's because your body responds to low oxygen levels by making more red blood cells to give you the oxygen you need.

Pregnancy can cause a low hematocrit. That's because your body has more fluid than usual during pregnancy, which decreases the percentage that's made of red blood cells.
To learn what your test results mean, talk with your provider.

MCV (Mean Corpuscular Volume)

What is an MCV blood test?

MCV stands for mean corpuscular volume. An MCV blood test measures the average size of your red blood cells.

Red blood cells carry oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy. If your red blood cells are too small or too large, it could be a sign of a blood disorder such as anemia, a lack of certain vitamins, or other medical conditions.
Other names: CBC with differential, Mean Cell Volume, Mean Corpuscular Volume

What is it used for?

An MCV blood test is often part of a complete blood count (CBC), a group of tests that measures the number and type of cells in your blood. It is used to check your general health.
An MCV test may also be used with other tests to help diagnose or monitor certain blood disorders, including anemia. There are several types of anemia, and each type has a different effect on the size, shape, and/or quality of your red blood cells. An MCV test can help diagnose which type of anemia you have.

Why do I need an MCV blood test?

Your health care provider may order a complete blood count, which includes an MCV test, as part of your routine checkup. You may also have the test if you have a chronic (long-lasting) condition that could lead to anemia or if you have symptoms of anemia, which could include:

- Shortness of breath
- Weakness or fatigue
- Headache
- Dizziness
- Arrhythmia (a problem with the rate or rhythm of your heartbeat)

What happens during an MCV blood test?

During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

You don't need any special preparations for an MCV blood test. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

An MCV test alone cannot diagnose any disease. Your provider will use the results of your MCV, other test results, and your medical history to make a diagnosis.

If your results show that your red blood cells are smaller than normal, it may be a sign of:

- Certain types of anemia, including iron deficiency anemia, the most common form of anemia. It happens when you don't have enough iron in your body.
- Thalassemia, which is a group of blood disorders that are inherited (passed down through families). These disorders cause your body to make fewer healthy red blood cells and less hemoglobin. This can cause severe anemia.

If your results show that your red blood cells are larger than normal, it may be a sign of:

- Pernicious anemia, which may be caused by:
    - A lack of vitamin B12
    - A disease that affects your body's ability to use vitamin B12, such as certain autoimmune diseases, celiac disease, or Crohn's disease
- Anemia caused by a lack of folic acid
- Liver disease

It's also possible to have anemia with **a normal MCV**. This may happen if anemia is caused by conditions, such as:

- A sudden loss of blood
- Kidney failure
- Aplastic anemia (uncommon)

If your MCV levels are not in the normal range, it doesn't always mean you have a medical problem that needs treatment. Your age, diet, activity level, medicines, menstrual period, and other conditions can affect the test results. Talk with your provider to learn what your results mean.

Is there anything else I need to know about an MCV blood test?

If your provider thinks you may have anemia or another blood disorder, you may have other red blood cell tests with an MCV. These tests may include a red blood cell count and measurements of hemoglobin. All together, these tests are called red blood cell indices.

Red Blood Cell (RBC) Indices

What are red blood cell (RBC) indices?

Red blood cell (RBC) indices measure your red blood cells' size, shape, and quality. Red blood cells are also known as erythrocytes. They are made in your bone marrow (the spongy tissue inside your large bones). They contain hemoglobin, an iron-rich protein in your red blood cells that carries oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy.

Knowing the size and shape of your red blood cells can help your provider determine if you have a certain type of anemia, a condition in which your body does not make enough healthy red blood cells. There are four types of red blood cell indices:

- **Mean corpuscular volume (MCV)**, which measures the average size of your red blood cells.

- **Mean corpuscular hemoglobin (MCH)**, which measures the average amount of hemoglobin in a single red blood cell.
- **Mean corpuscular hemoglobin concentration (MCHC)**, which measures how concentrated (close together) the hemoglobin is in your red blood cells. It also includes a calculation of the size and volume of your red blood cells.
- **Red cell distribution width (RDW)**, which measures differences in the volume and size of your red blood cells. Healthy red blood cells are usually about the same size.

If one or more of these indices are not normal, it may mean you have some type of anemia.

Other names: erythrocyte indices

What are they used for?

Red blood cell (RBC) indices are part of a complete blood count, a group of tests that measures the number and type of cells in your blood. The results of RBC indices are used to diagnose different types of anemia. There are several types of anemia, and each type has a different effect on the size, shape, and/or quality of red blood cells.

Why do I need red blood cell indices testing?

You may get this test as part of a complete blood count, which is often included in a routine checkup. You may also need this test if you have symptoms of anemia, which may include:

- Shortness of breath
- Weakness or fatigue
- Headache
- Dizziness
- Arrhythmia (a problem with the rate or rhythm of your heartbeat)
- Pale skin
- Cold hands and feet

What happens during a red blood cell indices test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This test usually takes less than five minutes.

Will I need to do anything to prepare for these tests?

You don't need any special preparations for a red blood cell (RBC) indices test.

Are there any risks to these tests?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

You will get results for each of the indices. Abnormal results may include one or more of the following:

Mean corpuscular volume (MCV)

If your red blood cells are **smaller than normal**, it may mean you have:

- Iron deficiency anemia, the most common form of anemia. It happens when you don't have enough iron in your body.
- Thalassemia, which is a group of blood disorders that are inherited (passed down through families). These disorders cause your body to make fewer healthy red blood cells and less hemoglobin. This can cause severe anemia.

If your red blood cells are **larger than normal**, it may mean you have:

- Anemia caused by a vitamin B deficiency
- Liver disease

Mean corpuscular hemoglobin (MCH)

If the amount of hemoglobin is **lower than normal**, it may mean you have:

- Iron deficiency anemia

If the amount of hemoglobin is **higher than normal**, it may mean you have:

- Anemia caused by a vitamin B deficiency

Mean corpuscular hemoglobin concentration (MCHC)

If the average amount of hemoglobin is **lower than normal**, it may mean you have:

- Iron deficiency anemia
- Thalassemia

If the average amount of hemoglobin is **higher than normal**, it may mean you have:

- Hemolytic anemia, a type of anemia that happens when red blood cells are destroyed faster than they can be replaced.
- Hereditary spherocytosis, a rare genetic disorder that causes anemia and gallstones

Red cell distribution width (RDW)

- If your **results were normal**, it means your red blood cells are normal in size and are all about the same size.
- If your **results were not normal**, it means there are differences in the size of your red blood cells.

This measurement is not enough for your provider to make a diagnosis, so RDW results are usually combined with the results of other indices and other blood tests. This combination of results can help confirm a diagnosis.

Talk to your health care provider if you have questions about your results. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand your results.

Erythrocyte Sedimentation Rate (ESR)

What is an erythrocyte sedimentation rate (ESR)?

Erythrocytes are red blood cells. The sedimentation rate is the time it takes for your red blood cells to settle at the bottom of a test tube. An erythrocyte sedimentation rate (ESR) is a blood test that can show if you have inflammation in your body. Inflammation is your immune system's response to injury, infection, and many other conditions, including immune system disorders, certain cancers, and blood disorders.
To do an ESR test, a sample of your blood is sent to a lab. A health care professional places the sample in a tall, thin test tube and measures how quickly your red blood cells settle or sink to the bottom of the tube. Typically, red blood cells sink slowly. But inflammation makes red blood cells stick together in clumps. These clumps of cells are heavier than single cells, so they sink faster.

If an ESR test shows that your red blood cells sink faster than expected, it may mean you have a medical condition causing inflammation. How quickly the red blood cells in your sample fall to the bottom of the test tube is a sign of how much inflammation you have. Faster ESR rates mean higher levels of inflammation. But an ESR test alone cannot diagnose which condition is causing the inflammation.

Other names: ESR, SED rate sedimentation rate; Westergren sedimentation rate

What is it used for?

An ESR test can be used with other tests to help diagnose and monitor conditions that cause inflammation. Many conditions cause inflammation, including arthritis, vasculitis, infections, and inflammatory bowel diseases (ulcerative colitis and Crohn's disease). An ESR may also be used to monitor an existing condition.

Why do I need an ESR?

Your health care provider may order an ESR if you have symptoms of a condition that causes inflammation. Your symptoms will depend on the condition you may have, but they may include:

- Headaches
- Unexplained fever
- Unexplained weight loss
- Joint stiffness
- Neck or shoulder pain
- Loss of appetite
- Anemia

What happens during an ESR?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for an ESR?

You don't need any special preparations for this test. But if your provider ordered other tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

Are there any risks to the test?

There is very little risk to having an ESR. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

Your provider will use your ESR test results, medical history, symptoms, and other test results to make a diagnosis. An ESR test alone cannot diagnose conditions that cause inflammation.

A high ESR test result means your red blood cells sank faster than normal. This may be from a condition that causes inflammation, such as:

- Giant cell arteritis, inflammation (swelling) of your arteries, usually in your scalp, neck, and arms
- Arthritis
- Systemic vasculitis, inflammation of your blood vessel walls, which can affect any of your organs
- Polymyalgia rheumatica
- Inflammatory bowel disease
- Kidney disease
- Infections
- Rheumatoid arthritis and other autoimmune diseases
- Heart disease
- Certain cancers

A low ESR test result means your red blood cells sank slower than normal. This may be caused by conditions such as:

- A blood disorder, such as:
  - Polycythemia, an increase in your red blood cells, which causes your blood to thicken
  - Sickle cell disease (SCD)
  - Leukocytosis, a very high white blood cell count (WBC)
- Heart failure
- Certain kidney and liver problems

If your ESR results are not normal, it doesn't always mean you have a medical condition that needs treatment. Normal ESR levels will vary, depending on your age and sex. Also, pregnancy, your menstrual cycle, aging, obesity, drinking alcohol regularly, and exercise can affect ESR results. Certain medicines and supplements may also affect your results, so be sure to tell your provider about any medicines or supplements you are taking.
Learn more about laboratory tests, reference ranges, and understanding results.

Is there anything else I need to know about an ESR?

Because an ESR can't diagnose a specific disease, your provider may order other tests at the same time. Also, it's possible to have a condition that causes inflammation and still have a normal ESR result. A C-reactive protein (CRP) test is commonly done with an ESR to provide more information.

Blood Glucose Test

What is a blood glucose test?

A blood glucose test measures the glucose levels in your blood. Glucose is a type of sugar. It is your body's main source of energy. It comes from the food you eat. A hormone called insulin

helps move glucose from your bloodstream into your cells for use as energy. If you have too much or too little glucose in your blood, it may be a sign of a serious medical condition.
High blood glucose levels (hyperglycemia) may be a sign of prediabetes or diabetes. **Prediabetes** is a condition in which your blood glucose levels are higher than what is healthy for you, but not high enough to be considered diabetes. If you have **diabetes**, your body can't make insulin or can't use it as well as it should, or both. Too much glucose stays in your blood and doesn't reach your cells. This can cause glucose levels to get too high. If blood glucose isn't controlled, it can lead to serious, long-term health conditions, such as heart disease and nerve problems.
High blood glucose may also be caused by other conditions that can affect insulin or glucose levels in your blood, such as problems with your pancreas or adrenal glands, and side effects from certain medicines.
Low blood glucose levels (hypoglycemia) is often caused by certain diabetes medicines for type 1 diabetes or type 2 diabetes, including insulin. In people without diabetes, low blood glucose is much less common. It may be caused by certain medicines and conditions, such as some kidney or liver diseases. Without treatment, severe low blood glucose can lead to major health problems, including seizures and brain damage.
Several types of blood tests check blood glucose levels. Your health care provider will choose the test that's right for you.

Other names: blood sugar, self-monitoring of blood glucose (SMBG), fasting plasma glucose (FPG), fasting blood sugar (FBS), fasting blood glucose (FBG), random blood sugar, glucose challenge test, oral glucose tolerance test (OGTT)

What is it used for?

A blood glucose test may be used:

- As part of a routine checkup to screen for prediabetes and diabetes
- To help diagnose the cause of symptoms that may mean your blood glucose is too high or too low
- To monitor the side effects of certain long-term medicines that may affect blood glucose

Why do I need a blood glucose test?

Your provider may order a blood glucose test as part of your routine checkup. Routine blood tests that include a glucose test include a basic metabolic panel (BMP) and a comprehensive metabolic panel (CMP).
The most common reasons for a blood glucose test are to screen for and monitor prediabetes and diabetes. Your provider may order a blood glucose test if:

- **You have a high risk for developing type 2 diabetes.** Your risk may be high if you:
  - Are over 35 years old. The American Diabetes Association recommends diabetes screening for prediabetes and diabetes beginning at age 35 years and older.
  - Have prediabetes.
  - Are overweight or have obesity.
  - Have a family history of diabetes.
  - Have high blood pressure or heart disease.
  - Have ever had gestational diabetes (diabetes that develops during pregnancy) or given birth to a baby who weighed over 9 pounds.
  - Exercise less than three times a week.
  - Have non-alcoholic fatty liver disease (NAFLD).
  - Are an African American, Hispanic or Latino, American Indian, or an Alaska Native.

Some Pacific Islander people and Asian American people also have a higher risk for developing type 2 diabetes.

- **You have symptoms of high or low glucose levels**, which could mean you have diabetes or another condition that affects blood glucose levels.

  Symptoms of **high blood glucose levels** include:
  - Increased thirst
  - Frequent urination (peeing)
  - Blurred vision
  - Fatigue
  - Sores that are slow to heal
  - Losing weight without trying
  - Numbness or tingling in your feet or hands

  Symptoms of **low blood glucose levels** include:
  - Feeling shaky or jittery
  - Hunger
  - Fatigue
  - Feeling dizzy, confused, or irritable
  - Headache
  - A fast heartbeat or arrhythmia (a problem with the rate or rhythm of your heartbeat)
  - Having trouble seeing or speaking clearly
  - Fainting or seizures

- **You have prediabetes or diabetes.** Your provider may schedule blood glucose tests to monitor your condition. You may also monitor your blood glucose at home.
- **You are pregnant.** You will likely have a blood glucose test between the 24th and 28th week of your pregnancy to check for gestational diabetes. If you have a high risk for gestational diabetes, your provider may test you earlier.
- **You take long-term medicines, other than diabetes medicines, that may affect blood glucose levels.** Examples of medicines that may cause high blood glucose in some people include certain types of:

  - Steroids
  - Antipsychotics
  - Beta-blockers
  - Statins

  Examples of medicines that may cause low blood glucose in some people include certain types of:
  - ACE inhibitors
  - Antibiotics called quinolones
  - Medicines to treat malaria

What happens during a blood glucose test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

There are several types of blood glucose tests:

- A **fasting blood glucose test** measures your glucose after you fast (not eating or drinking, except for water) for at least 8 hours. It is often used to:
  - Screen for or monitor prediabetes or diabetes
  - Help diagnose or monitor other conditions that affect blood glucose
  - Check for side effects from medications that can affect blood glucose
- A **random blood glucose test** can be done at any time, even after eating. If you have symptoms of diabetes, your provider may use this test first. That's because it can be done right away.
- A **glucose challenge test** may be used to screen for gestational diabetes if you are pregnant. You don't need to fast for this test. You will drink a sugary drink that contains glucose. A blood sample is taken one hour later. If your blood glucose is too high, you will usually need an oral glucose tolerance test to confirm or rule out gestational diabetes.
- An **oral glucose tolerance test (OGTT)** is used to diagnose gestational diabetes. If you aren't pregnant, it's sometimes used to confirm a diagnosis of prediabetes or type 2 diabetes. You will need to fast for this test. A blood sample will be taken before you have a sugary drink. After you have the drink, more blood samples will be taken, usually about every hour for the next 2 or 3 hours.
- A **hemoglobin A1C (HbA1c) test** provides information about your average blood glucose levels over the past 3 months. You don't need to fast for this test. It uses one blood sample. An A1C test can diagnose prediabetes and diabetes. Most people with diabetes have an A1C test at least twice a year to check how well they are managing their blood glucose levels.

Will I need to do anything to prepare for the test?

If your provider orders a fasting blood glucose test or an oral glucose tolerance test, you will need to fast for at least eight hours before the test. If your glucose test is part of a BMP or CMP test, you may need to fast. Other blood glucose tests don't require any special preparations. Ask your provider whether you need to fast before your glucose test.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. After an oral glucose tolerance test, you may feel light-headed. Your provider may suggest that you plan to have someone take you home.

What do the results mean?

The meaning of your test results will depend on which test you had, the results of other tests, and your medical history. Ask your provider what your test results say about your health.

In general:

If your results show higher than normal glucose levels, it may mean you have or are at risk for getting diabetes. Prediabetes and diabetes are the most common causes of high glucose levels, but other possible causes include:

- An overactive thyroid gland (hyperthyroidism)
- A problem with your pancreas, including pancreatitis
- Stress from surgery, very serious illness, or trauma
- A problem with your adrenal glands, including Cushing's syndrome and pheochromocytoma
- Certain medicines

If you have diabetes, glucose levels that are lower than normal for you may be caused by:

- Certain diabetes medicines and other medicines that may lower blood glucose
- Not eating enough, especially after taking diabetes medicine
- Being more physically active than usual

If you don't have diabetes, low blood glucose levels aren't common. If they happen, they may be caused by:

- Liver or kidney disease
- Underactive adrenal glands, for example, Addison disease
- Certain pituitary disorders
- An underactive thyroid gland (hypothyroidism)
- Malnutrition
- Certain medicines
- Alcohol use disorder (AUD)

Is there anything else I should know about a blood glucose test?

If you have diabetes, you may need to do blood glucose testing at home to help manage your blood glucose levels. Your provider will let you know how often you should test. There are two ways to check your blood glucose. Your provider can help you decide which one is best for you:

- **Blood glucose meters** require you to prick your finger with a small device called a lancet. Then, you apply a drop of blood to a test strip and insert it into a small electronic glucose meter, which measures the glucose in your blood.
- **Continuous glucose monitors (CGM)** usually use a tiny sensor placed under the skin of your arm or belly. Depending on the type of CGM you use, the sensor stays in place from a week to several months. It automatically estimates your glucose level every few minutes and tracks it so you can see what your blood glucose level is at any time.

Diabetes Tests

What are diabetes tests?

Diabetes, also known as diabetes mellitus, is a disease that affects how your body uses glucose (blood sugar). Glucose is your body's main source of energy. A hormone (a chemical messenger in your bloodstream that controls the actions of certain cells or organs) called insulin helps move glucose from your bloodstream into your cells.

If you have diabetes, your body can't make insulin or can't use it as well as it should, or both. Too much glucose stays in your blood and doesn't reach your cells. This can cause glucose levels to get too high, which can lead to serious health conditions. These may include heart disease, nerve damage, eye problems, and kidney disease.
Your blood glucose can also drop below what is healthy for you. If it drops very low, you could lose consciousness or have a seizure.
Diabetes tests most often measure glucose levels in your blood to monitor your blood glucose level or see if you have or are at risk for diabetes. Urine tests for diabetes are not used as much since they are not as accurate, and you would need a blood test to diagnose diabetes.

Other names: blood glucose, fasting plasma glucose, FPG, oral glucose tolerance test, OGTT, glucose screening test, glucose in urine test, hemoglobin A1c, random blood sugar, blood sugar, fasting blood sugar (FBS)

What are they used for?

Diabetes tests are used to screen, monitor, and/or diagnose the following:

- **Type 1 diabetes.** If you have type 1 diabetes, your body makes little or no insulin. This disease happens when your immune system attacks and destroys cells that produce insulin. It can develop at any age but most often starts in childhood. People with type 1 diabetes must take daily doses of insulin, either by injection or a special pump.
- **Type 2 diabetes.** This is the most common form of diabetes. If you have type 2 diabetes, your body may still be able to make insulin, but your cells don't respond well to insulin and can't easily take up enough glucose from your blood. Type 2 diabetes may be caused by your genes and lifestyle factors, such as being overweight or having obesity. The disease most often occurs in adulthood but is becoming more common in children and teens.
- **Gestational diabetes.** This is a form of diabetes that develops during pregnancy.
- **Prediabetes.** This is a condition in which your blood glucose levels are higher than what is healthy for you but not high enough to be considered diabetes. But having prediabetes raises your risk of getting type 2 diabetes later on.

Why do I need a diabetes test?

You may need testing to monitor your blood glucose level or if you have symptoms of diabetes, such as:

- Increased thirst
- Frequent urination
- Increased hunger
- Fatigue
- Blurred vision
- Unexplained weight loss
- Sores that are slow to heal
- Numbness or tingling in your feet

Symptoms of type 1 diabetes usually come on quickly and can be severe. Symptoms of type 2 diabetes and prediabetes often develop slowly, even over years.

Gestational diabetes doesn't usually cause symptoms in the early stages of your pregnancy, but you will most likely be screened for the condition between 24 and 28 weeks of pregnancy. If testing shows your glucose levels are too high, you will be tested again to confirm the diagnosis.

You may also need testing if you have certain risk factors for type 2 diabetes. You may be at higher risk for developing it if you:

- Are over 35 years old. The American Diabetes Association recommends diabetes screening for prediabetes and diabetes beginning at age 35 years and older.
- Have prediabetes.
- Are overweight or have obesity.
- Have a family history of diabetes.
- Have high blood pressure or heart disease.
- Previously had gestational diabetes.
- Exercise less than three times a week.
- Have non-alcoholic fatty liver disease (NAFLD).
- Are an African American, Hispanic or Latino, American Indian, or an Alaska Native.

What happens during a diabetes test?

There are several ways to screen for, monitor, and diagnose diabetes. Most tests involve measuring your blood glucose levels.

To get a blood sample, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

The different types of **glucose blood tests** include:

- **Blood glucose test**, also known as fasting blood glucose. Before the test, you will need to fast (not eat or drink) for eight hours before the test. This test is usually done in the morning and is often used as a screening test for diabetes. It may be repeated to confirm a diagnosis.
- **Oral glucose tolerance test (OGTT).** This test also requires fasting. When you arrive for your test, a blood sample will be taken. You will then drink a sugary liquid that contains glucose. About two hours later, another blood sample will be taken. This test tells your health care provider how well your body processes sugar. This test may be used to test for gestational diabetes. If you are having this test done during pregnancy, you will have your blood drawn every 30 minutes for two to three hours.
- **Glucose challenge test**, also known as the glucose screening test. This test may be used to test for gestational diabetes. You will drink a sugary liquid that contains glucose. A blood sample will be taken one hour later. You do not need to fast for this test. If your blood glucose is too high, your provider may order an oral glucose tolerance test.
- **Random blood sugar.** This test can be taken at any time. No fasting is required. This test may be used if you have symptoms of diabetes.
- **Hemoglobin A1c (HbA1c).** This test measures the average amount of glucose attached to hemoglobin over the past three months. Hemoglobin is the part of your red blood cells that carries oxygen from your lungs to the rest of your body. No fasting is required for this test.

Glucose may also be measured in your urine.
Urine tests are not often used to diagnose or monitor diabetes since blood tests are more accurate. But a urine test may show if you are at risk for getting the disease. If your glucose in urine levels are too high, then you will probably need a blood test to confirm the diagnosis.

For a glucose in urine test, your provider may recommend an at-home test kit. The kit will include a test strip that you hold under your stream of urine. The test strip will change colors to show different levels of glucose.

Will I need to do anything to prepare for this test?

You will need to fast (not eat or drink) for a blood glucose and an oral glucose tolerance test.

You don't need any special preparations for a random blood sugar test, glucose challenge test, hemoglobin A1c test, or glucose in urine test.

Are there any risks to this test?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a urine test.

What do the results mean?

Depending on the type of test or tests you had, your results may show one of the following:

- **Your glucose is at a healthy level for you.** This means you probably are not at risk for or do not have diabetes.
- **Prediabetes.**This means you have glucose levels that are higher than what is healthy for you and may be at risk of getting diabetes.
- **Type 1 or type 2 diabetes.**
- **Gestational diabetes.**

If you have been diagnosed with type 1 diabetes, talk to your provider about how to best manage the disease. There is no cure for type 1 diabetes, but it may be controlled with regular glucose monitoring and taking insulin.

If you've been diagnosed with prediabetes or type 2 diabetes, your condition may be managed or even reversed by taking diabetes medicines and making lifestyle changes. These include eating a healthy diet, losing weight, and increasing exercise.
If you've been diagnosed with gestational diabetes, you may be able to lower your glucose levels by eating a healthy diet and getting regular exercise. But be sure to talk to your provider about treatment options. Most of the time, gestational diabetes goes away after giving birth. But you will be at higher risk for developing type 2 diabetes in the future.

If you have other questions about your diabetes diagnosis or treatment, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

Is there anything else I need to know about diabetes testing?

If you've been diagnosed with type 1 diabetes, you will need to monitor your blood glucose levels daily, often several times a day. Your provider can recommend a kit you can use at home. Most kits include a lancet, a device that pricks your finger. You will use this to collect a drop of blood for testing. There are some newer kits available that don't require pricking your finger. If you are pregnant and have gestational diabetes, you may also need to monitor your glucose levels in this way.

People with type 2 diabetes must also check their blood sugar regularly. If you have type 2 diabetes, talk to your provider about how often it should be checked.

People with type 2 diabetes may also need to have their insulin levels checked regularly. Insulin plays a key role in keeping glucose at the right levels. An insulin in blood test is done at a provider's office.

Hemoglobin A1C (HbA1c) Test

What is a hemoglobin A1C (HbA1C) test?

A hemoglobin A1C (HbA1C) test is a blood test that shows your average level of blood glucose, also called blood sugar, over the past two to three months.
Glucose is a type of sugar in your blood that comes from the foods you eat. Your cells use glucose for energy. A hormone called insulin helps glucose get into your cells. If you have diabetes your body doesn't make enough insulin, or your cells don't use it well. As a result, glucose can't get into your cells, so your blood glucose levels increase.
Glucose in your blood sticks to hemoglobin, a protein in your red blood cells. As your blood glucose levels increase, more of your hemoglobin will be coated with glucose. An A1C test measures the percentage of your red blood cells that have glucose-coated hemoglobin.

An A1C test can show your average glucose level for the past three months because:

- Glucose sticks to hemoglobin for as long as the red blood cells are alive.
- Red blood cells live about three months.

High A1C levels are a sign of high blood glucose from diabetes. Diabetes can cause serious health problems, including heart disease, kidney disease, and nerve damage. But with treatment and lifestyle changes, you can control your blood glucose levels.
Other names: HbA1C, A1C, glycohemoglobin, glycated hemoglobin, glycosylated hemoglobin

What is it used for?

Your blood glucose level changes all the time. If you're checking it at home, you are measuring what your blood glucose level is at that time. An A1C test gives you an average level of your blood glucose level over the past two to three months.

An A1C test may be used to screen for or diagnose:

- **Type 2 diabetes.** With type 2 diabetes your blood glucose gets too high because your body doesn't make enough insulin or doesn't use insulin well. The glucose then stays in your blood and not enough goes into your cells.
- **Prediabetes.** Prediabetes means that your blood glucose levels are higher than normal, but not high enough to be diagnosed as diabetes. Lifestyle changes, such as healthy eating and exercise, may help delay or prevent prediabetes from becoming type 2 diabetes.

If you have diabetes or prediabetes, an A1C test can help monitor your condition and check how well you've been able to control your blood glucose levels.

The A1C test isn't used to diagnose type 1 diabetes but may be used for monitoring your blood glucose levels.

Why do I need an A1C test?

The Centers for Disease Control (CDC) recommends A1C testing for diabetes and prediabetes if:

- **You are over age 45.**
    - If your results are normal, your provider will tell you how often you should be tested based on your age and risk factors.
    - If your results show you have prediabetes, you will usually need to be tested every 1 to 2 years. Ask your provider how often to get tested and what you can do to reduce your risk of developing diabetes.
    - If your results show you have diabetes, you should get an A1C test at least twice a year to monitor your condition and treatment.
- **You are under 45 and are more likely to develop diabetes because you:**
    - Have prediabetes.
    - Are overweight or have obesity.
    - Have a parent or sibling with type 2 diabetes.
    - Have high blood pressure or high cholesterol levels.
    - Have heart disease or have had a stroke.
    - Are physically active less than 3 times a week.
    - Have had gestational diabetes (diabetes during pregnancy) or given birth to a baby over 9 pounds.
    - Are African American, Hispanic or Latino, American Indian, or an Alaska Native person. Some Pacific Islander and Asian American people also have a higher risk of developing diabetes.
    - Have polycystic ovarian syndrome (PCOS).

You may also need an A1C test if you have symptoms of diabetes, such as:

- Feeling very thirsty
- Urinating (peeing) a lot
- Losing weight without trying
- Feeling very hungry
- Blurred vision
- Numb or tingling hands or feet
- Fatigue
- Dry skin
- Sores that heal slowly
- Having more infections than usual

What happens during an A1C test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

Certain medicines such as opioids and some HIV medicines may affect your results, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. Also, let your health care provider know if you are pregnant, since this may affect your results.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

A1C results tell you what percentage of your hemoglobin is coated with glucose. The ranges are just a guide to what is normal. What's normal for you depends on your health, age, and other factors. Ask your provider what A1C percentage is healthy for you.

To diagnose diabetes or prediabetes, the percentages commonly used are:

- **Normal:** A1C below 5.7%
- **Prediabetes:** A1C between 5.7% and 6.4%
- **Diabetes:** A1C of 6.5% or higher

Providers often use more than one test to diagnose diabetes. So, if your test result was higher than normal, you may have another A1C test or a different type of diabetes test, usually either a fasting blood glucose test or an oral glucose tolerance test (OGTT).
If your A1C test was done to monitor your diabetes, talk with your provider about what your test results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

Is there anything else I need to know about an A1C test?

The A1C test is not used to diagnose gestational diabetes. Also, if you have a condition that affects your red blood cells, such as anemia or another type of blood disorder, an A1C test may not be accurate for diagnosing diabetes. Kidney failure and liver disease can also affect A1C results. In these cases, your provider may recommend different tests to diagnose diabetes and prediabetes.

Insulin in Blood

What is an insulin in blood test?

An insulin in blood test measures the amount of insulin in a sample of your blood. Insulin is a hormone that your pancreas makes. It helps move blood glucose (blood sugar) from your bloodstream into your cells where it's used for energy. Glucose comes from many foods you eat. It's your body's main source of energy.
Normally, insulin and blood glucose levels rise and fall together:

- Blood glucose levels increase after you eat.
- When blood glucose rises, your pancreas releases more insulin into your blood.
- The insulin lets glucose get into your cells, which lowers your blood glucose level.
- When your blood glucose level returns to a range that's normal for you, your insulin levels decrease, too.

Serious problems can develop if your pancreas doesn't make the right amount of insulin at the right time. For example:

- **If your pancreas makes too much insulin**, you may have too little glucose in your blood. This is called hypoglycemia. If blood glucose levels drop to a very low level (severe hypoglycemia), your brain cells may not get enough glucose to work properly. This is a serious condition that needs medical treatment right away.
- **If your pancreas makes too little insulin**, glucose can't get into your cells from your blood. The glucose builds up in your bloodstream until your blood glucose level is too high. This is called hyperglycemia. If your blood glucose levels stay high over time, you could develop type 2 diabetes, which can cause serious complications in your eyes, heart, and other parts of your body.

If you have signs or symptoms of a health condition related to insulin, an insulin in blood test can help check how much insulin your pancreas is making.

Other names: fasting insulin, insulin serum, total and free insulin

What is it used for?

An insulin in blood test may be used with other tests to help:

- **Find out the cause of hypoglycemia (low blood glucose).** This is the main reason for doing insulin testing.
- **Diagnose insulin resistance.** With insulin resistance, your cells don't respond well to insulin and can't easily take in glucose from your blood. Your pancreas may make more insulin to help glucose get into your cells. This can keep your blood glucose in a healthy range for a while. But over time, your pancreas may wear out and stop making enough insulin. This can lead to prediabetes, which means your blood glucose levels are higher-than-normal, but not high enough to be diabetes. If blood glucose levels keep increasing, you can develop type 2 diabetes.
- **Guide treatment decisions for type 2 diabetes.** An insulin test may be used to decide if a person with type 2 diabetes needs to take insulin as part of their diabetes treatment.
- **To monitor how well pancreatic islet cell transplantation surgery is working.** This surgery is used in people who have type 1 diabetes. Type 1 diabetes is an autoimmune disease that destroys the special islet cells in the pancreas that make insulin.

Why do I need an insulin in blood test?

You may need an insulin in blood test if you:

- Had a blood glucose test that showed you have low blood glucose levels.
- Have symptoms of hypoglycemia (low blood glucose). Symptoms of mild to moderate hypoglycemia include:
  - Sweating
  - Feeling shaky or jittery
  - Arrythmia (a problem with the rate or rhythm of your heartbeat)
  - Confusion
  - Dizziness
  - Headache
  - Hunger

Severe hypoglycemia can cause fainting and seizures. This serious condition needs medical treatment right way.

- Have insulin resistance or have a high risk for developing it. Your risk for insulin resistance is higher if you have:
    - Prediabetes
    - Type 2 diabetes
    - Polycystic ovary syndrome (PCOS)
    - Heart disease
    - Acanthosis nigricans (dark, thick, velvety skin around the neck or armpits)
- Had surgery on your pancreas:
    - To transplant the part of the pancreas that makes insulin (islet cell transplantation). You may need insulin testing to see if the transplant is working properly.
    - To remove an insulinoma. This is a tumor in the pancreas that makes too much insulin and causes low blood glucose. After surgery you may need insulin testing to check whether any tumor tissue is left and to see if the tumor has come back. Insulinomas are uncommon and usually aren't cancer.

What happens during an insulin in blood test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

Will I need to do anything to prepare for the test?

Your health care provider will tell you how to prepare for an insulin in blood test. You will probably need to fast (not eat or drink) for 8 to 12 hours before the test. If you take biotin supplements (vitamin B7), or supplements that include biotin, you'll need to stop taking them for at least a day before your test. But don't stop taking any medicines without talking with your provider first.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

To understand what your insulin test results mean, your provider will consider your medical history and results of other tests, including a blood glucose test. For example:

- If your insulin level is high and your blood glucose is normal or a little above normal for you, you may have insulin resistance.
- If your insulin level is high or normal and your blood glucose is low for you, you may have hypoglycemia from too much insulin. Possible causes include:
    - A pancreatic tumor (insulinoma)
    - Cushing's syndrome
    - Taking too much insulin for diabetes

- If your insulin level is low and your blood glucose is high for you, it may mean that your pancreas can't make enough insulin. Possible causes include type 1 diabetes and pancreatitis.

There are other possible causes of abnormal levels of insulin and glucose. Ask your provider to explain what your test results say about your health.

Is there anything else I should know about an insulin in blood test?

An insulin in blood test is often done with a C-peptide test. Your pancreas releases equal amounts of insulin and C-peptide into your bloodstream at the same time. C-peptide doesn't affect your blood glucose levels, but it stays in your blood longer than insulin. So, measuring C-peptide provides a more accurate way to find out how much insulin your pancreas is making.

Cholesterol Levels

What is a cholesterol test?

A cholesterol test is a blood test that measures the amount of cholesterol and triglycerides (a type of fat) in your blood. Cholesterol is a waxy, fat-like substance that's found in your blood and every cell of your body. Your body needs some cholesterol to make hormones, vitamin D, and substances that help you digest foods.

Your liver makes all the cholesterol your body needs and removes excess amounts. Cholesterol is also found in foods from animal sources, such as meat, egg yolks, poultry, and dairy products. Foods high in dietary fat can increase the cholesterol in your blood. If there's too much cholesterol in your blood, your liver can't remove it all.

There are two main types of cholesterol: LDL (low-density lipoprotein), or "bad" cholesterol, and HDL (high-density lipoprotein), or "good" cholesterol.

Too much LDL cholesterol in your blood increases your risk for coronary artery disease and other heart diseases. High LDL levels can cause the buildup of a sticky substance called plaque in your arteries. This buildup of plaque is known as atherosclerosis. Over time, plaque can narrow your arteries or fully block them. When this happens, parts of your body may not get enough blood:

- If the blood flow to your heart is blocked, it can cause a heart attack.
- If the blood flow to your brain is blocked, it can cause a stroke.
- If the blood flow to your arms or legs is blocked, it can cause peripheral artery disease.

Other names for a cholesterol test: Lipid profile, Lipid panel

What is it used for?

A cholesterol test gives you and your health care provider important information about your risk of developing heart disease. If your test shows you have high cholesterol, you can take steps to lower it. This may decrease your risk of developing heart problems in the future.

A cholesterol test measures your:

- **LDL level.** LDL ("bad") cholesterol is the main source of blockages in the arteries.
- **HDL level.** HDL ("good') cholesterol helps get rid of "bad" LDL cholesterol. A higher HDL level may help reduce your risk of heart attack or stroke.
- **Total cholesterol.** This is a measure of the total amount of cholesterol in your blood. It includes HDL and LDL cholesterol.

- **Triglyceride level.** Triglycerides are a type of fat found in your blood. High levels of triglycerides may increase the risk of heart disease, especially in women.

There is another type of cholesterol called VLDL (very low-density lipoprotein). Some people also call VLDL a "bad" cholesterol because it contributes to the buildup of plaque in your arteries. But VLDL and LDL are different; VLDL mainly carries triglycerides, and LDL mainly carries cholesterol. VLDL isn't usually included in routine cholesterol tests because it's difficult to measure. Because VLDL contains a certain percentage of triglycerides, a lab can use your triglycerides level to estimate your VLDL level.

Why do I need a cholesterol test?

Your provider may order a cholesterol test as part of a routine exam. When and how often you should get a cholesterol test depends on your age, risk factors, and family history. The general recommendations are:

For people who are age 19 or younger:

- The first test should be between ages 9 to 11
- Children should have the test again every 5 years
- Some children may have this test starting at age 2 if there is a family history of high cholesterol, heart attack, or stroke

For people who are ages 20 to 65:

- Younger adults should have the test every 5 years
- Men ages 45 to 65 and women ages 55 to 65 should have it every 1 to 2 years

For people older than 65:

- They should be tested every year

You may also have a cholesterol test more often if you are at high risk of heart problems because of:

- A family history of heart disease
- High blood pressure
- Type 2 diabetes
- Smoking
- Excess weight or obesity
- Lack of physical activity
- A diet high in saturated fat

What happens during a cholesterol test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

You may be able to use an at-home kit to check your cholesterol levels. Your kit will include a device to prick your finger to collect a drop of blood for testing. Be sure to follow the kit instructions carefully. Also, be sure to tell your provider if your at-home test shows that your total cholesterol level is higher than 200 mg/dl.

Will I need to do anything to prepare for the test?

You may need to fast (not eat or drink) for 9 to 12 hours before your blood cholesterol test. That's why the tests are often done in the morning. Your provider will let you know if you need to fast and if there are any other special instructions.

Are there any risks to the test?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

Cholesterol is usually measured in milligrams (mg) of cholesterol per deciliter (dL) of blood. The information below will help you understand what your test results mean. In general, low LDL levels and high HDL cholesterol levels are good for heart health.

Anyone age 19 or younger:

Type of Cholesterol	Healthy Level
Total Cholesterol	Less than 170 mg/dL
Non-HDL	Less than 120 mg/dL
LDL	Less than 110 mg/dL
HDL	More than 45mg/dL

Men age 20 or older:

Type of Cholesterol	Healthy Level
Total Cholesterol	Less than 200 mg/dL
Non-HDL	Less than 130 mg/dL
LDL	Less than 100 mg/dL
HDL	Greater than or equal to 60 mg/dL is best. Levels less than 40 mg/dL are considered low.

Women age 20 or older:

Type of Cholesterol	Healthy Level
Total Cholesterol	Less than 200 mg/dL
Non-HDL	Less than 130 mg/dL
LDL	Less than 100 mg/dL

The LDL listed on your results may say "calculated." This means that your LDL level is an estimate based on your total cholesterol, HDL, and triglycerides. Your LDL level may also be measured "directly" from your blood sample. Either way, you want your LDL number to be low.

A healthy cholesterol level for you may depend on your age, family history, lifestyle, and other risk factors for heart disease, such as high triglyceride levels. Your provider can explain what's right for you.

Is there anything else I need to know about my cholesterol levels?

High cholesterol can lead to heart disease, the number one cause of death in the United States. You can't change some risk factors for high cholesterol, such as age and your genes. But there are actions you can take to lower your LDL levels and reduce your risk, including:

- **Eat a healthy diet.** Reduce or avoid foods high in saturated fat and cholesterol to help lower the cholesterol levels in your blood.
- **Manage your weight.** Being overweight can increase your cholesterol and risk for heart disease.
- **Stay active.** Regular physical activity may help lower your LDL (bad) cholesterol levels and raise your HDL (good) cholesterol levels. It may also help you lose weight.
- **Quit smoking.** Smoking lowers your HDL (good) cholesterol, especially in women. Smoking also raises your LDL (bad) cholesterol.
- **Reduce stress.** Stress may raise levels of certain hormones such as corticosteroids. These can cause your body to make more cholesterol.
- **Avoid drinking too much alcohol.** Alcohol can raise your total cholesterol level.

Talk to your provider before making any major change in your diet or exercise routine.

Postprandial Blood Sugar

What is Postprandial Blood Sugar?

Postprandial blood sugar refers to the glucose level in your blood after consuming a meal. The term "postprandial" literally means "after eating". When you eat, carbohydrates are broken down into glucose, which enters the bloodstream. This rise in blood sugar triggers the pancreas to release insulin, a hormone that helps cells absorb glucose for energy or storage. In healthy individuals, blood sugar typically peaks about one hour after eating and returns to baseline within two to three hours.

Why is PPBS Important?

Monitoring PPBS is crucial for several reasons:

- Early detection of diabetes or prediabetes: Elevated post-meal glucose can indicate insulin resistance or impaired glucose metabolism.
- Assessing diabetes management: PPBS helps evaluate how well medications, diet, and lifestyle are controlling blood sugar.
- Preventing complications: Persistent high postprandial glucose is linked to cardiovascular risks and other diabetes-related complications.
- Understanding metabolic health: Tracking PPBS can reveal which foods or meal patterns cause spikes, helping improve insulin sensitivity and overall health.

Normal Postprandial Blood Sugar Levels

- For non-diabetic individuals, blood sugar should generally be below 140 mg/dL two hours after a meal.
- For people with diabetes, the American Diabetes Association recommends a target of under 180 mg/dL, though some may benefit from tighter control.
- Blood sugar levels are usually measured two hours after eating, which is considered the standard for PPBS testing.

How PPBS Differs from Fasting Blood Sugar

Fasting blood sugar measures glucose after 8–10 hours without food, providing a baseline level of glucose in a resting state.

PPBS measures the body's response to a meal, showing how effectively insulin manages the post-meal glucose rise.

Both tests complement each other and are often used alongside HbA1c, which reflects average blood sugar over 2–3 months.

Factors Affecting Postprandial Blood Sugar

Meal composition: High-carbohydrate meals cause larger spikes.

Physical activity: Exercise improves insulin sensitivity and lowers post-meal glucose.

Stress and hormones: Stress hormones can elevate blood sugar.

Medications: Insulin and oral diabetes medications influence PPBS levels.

Managing Postprandial Blood Sugar

Balanced diet: Include complex carbs, lean proteins, healthy fats, and fiber.

Regular exercise: Helps cells use glucose efficiently.

Monitor glucose: Track PPBS to identify patterns and adjust meals or medications.

Hydration and stress management: Adequate water intake and stress-reduction techniques support stable glucose levels.

Understanding and monitoring postprandial blood sugar is essential for maintaining metabolic health, preventing diabetes complications, and optimizing energy levels throughout the day.